



A level **playing** field

Think you could make better maps for *Quake III Arena*? Here's a crash course in level designing to get you started

Pick a game, any game; observe the intricate play between the rules that define it and the environment within which it is cast. Specifically note how the said rules and the way they are rendered, as defined by a game level, always harmonise, dance in unity.

Take *Pac-Man* for instance. We follow a lonely dot as it attempts to escape the ghosts, even as it consumes little pills of joy—sometimes eating, at times eaten, always running! Whether in 2D or in 3D, the backdrop of the game always remains the same: an increasingly complex maze that poses a challenge in proportion to the player's skill. If the gameplay is escapism, the environment is a maze, be it *Pac-Man* or *Half-Life*.

Or look at *Serious Sam*. Pity this guy who is stuck in the middle of a desert and is about to be grounded beneath the hooves of fifty stampeding bulls. That is, if he is not blown to constituent polygons by those headless-kamikaze freaks first. Gaze then at the cannon that he carries in his hand, the one that shoots cannonballs, the kind that can lay five dead in a

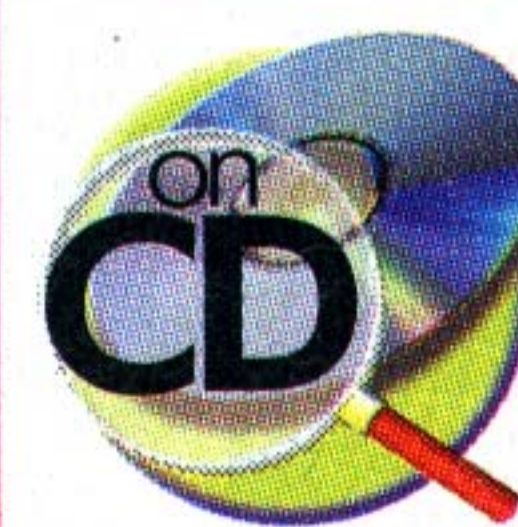
blow. Pity him no more. Besides, its not like Sam is stuck in a cramped room between narrow, claustrophobic corridors (like that poor *DOOM* guy), he has an entire desert to roam and run about. If the gameplay involves blowing up wave upon wave of indescribables, the levels must provide freedom to move about by being vast and open.

The same logic extends to other games. Where else can you loot, plunder, break into vehicles and be a public nuisance but in a metropolis? Thus *Grand Theft Auto 3* finds itself set in the sprawling muck that is Liberty City. A desire to kill your friends with alarming regularity will find yourself in closed, circular spaces with an assortment of weapons within easy reach—principles that drive games like *Quake III Arena* and *Unreal Tournament*. A good game is thus built on good levels and a good level, in turn, is one that emerges from the rules of the game; a square peg for a square hole. A perfect fit. Join us then as we examine the most important element of a game: level design. Take a look at the skills required to be a level designer as we introduce you to

some general design pointers, tools and resources that will help you take up the mantle of creating levels.

A part of the game

A game is little more than the sum of its parts. What separates a good game from a great one is the quality and skill with which those constituents are crafted. The engine forms the heart of any game. Analogous to a car, a game engine drives the rest of the package and is in turn composed of several elements. We thus have an AI (Artificial Intelligence) module that defines the behaviour of game characters (the level of aggression of the ghosts in *Pac-Man*, for example), an interface for sound and music in the game, a set of rules that define the limits to which physics dictates the game elements, amongst other things like scripting and event triggering.



Discreet's gmax along with companion resources
Find them in the Developer Tools section on the Mindware CD